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Abstract

What is my research about:

My research focuses on assessing and evaluating the current levels of radiological contamination in the environment of the United Kingdom (UK), specifically linked to the fallout from the Chernobyl Nuclear Power Plant (ChNPP) incident on April 26th, 1986. On May 1st, 1986, the radioactive cloud—containing various radionuclides—breached UK borders as it was carried by wind currents, with rainfall depositing contaminants across several regions.

To satisfy the technical requirements of the Cyber and Technical Computing component of the postgraduate research programme, I will be designing, developing, and deploying a Wireless Sensor Network (WSN), named *GammaNet*. This system will be used to monitor and evaluate beta and gamma radiation levels in areas identified through historical cloud movement and meteorological data from 1986. The aim is to collect real-time data for analysis and evaluation to assess current environmental conditions.

This research will investigate the integration of Internet of Things (IoT) technologies into a Wireless Sensor Network (WSN) to facilitate the capture and transmission of real-time environmental data for further analysis. I will also explore the implementation of a star network topology to enable efficient communication between nodes and examine the use of 433 MHz frequencies to support long-range data transmission.

In addition to the environmental monitoring component, my research will investigate the Chernobyl disaster in depth. This includes examining the events leading up to the incident at 01:23:40 AM, a step-by-step reconstruction of the accident, identification of those responsible, immediate post-incident actions, and a review of ongoing developments at the site up to the present day (2026).

Finally, I will undertake a series of smaller research projects throughout the course of my programme. Each project will contribute foundational data, methods, or insight, forming integral components that will feed into and support the final Field Research Project.

SOCK – Significant and Original Contribution to Knowledge

1 – MaRTI Research Project

The MaRTI project (Chapter 4, Subsection 4.1) offers a **significant and original contribution to knowledge** by addressing a critical but underexplored area in embedded systems and radiation research: the performance and survivability of widely-used, low-cost microcontrollers in radiation-rich environments.

Why is it significant:

1. **Real-world Relevance:** There is a growing interest in deploying low-cost, accessible technologies (like Arduino) in research satellites (CubeSats), high-altitude balloons, and even ground-based radiation monitoring systems. However, without data on how these devices handle radiation, designers are operating in the dark.
2. **Cost-Driven Innovation:** The research supports innovation in budget-constrained environments (e.g., education, small-scale space missions, citizen science), enabling more inclusive and scalable experimentation.
3. **Cross-Disciplinary Impact:** Findings will benefit multiple fields—space science, medical technology, nuclear engineering, and electronics—by providing a clearer understanding of failure mechanisms and tolerance thresholds in radiation-affected systems.

Why is it Original

1. **COTS Focus:** Unlike existing studies that primarily investigate radiation-hardened or space-grade components, MaRTI focuses specifically on **Commercial Off-The-Shelf (COTS)** microcontrollers. This shift in focus introduces new data and perspectives into the research landscape.
2. **Empirical Testing:** The project contributes **first-hand experimental results** on Arduino microcontrollers' behaviour under radiation—data that is currently sparse in academic literature.
3. **Practical Framework:** MaRTI is likely to propose **realistic recommendations or design adaptations** (e.g., shielding, redundancy,

watchdog timers) that can be implemented by developers using Arduino platforms in critical applications.

Summary:

In summary, this research fills a practical and scholarly void by generating new data, validating real-world use cases, and informing best practices for low-cost microcontroller use in environments where radiation is a risk factor. It bridges the gap between academic radiation research and the maker/engineering community, ultimately contributing both theoretical insight and actionable outcomes.

2 – Next Project

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Why is it significant:

We;rkjtng;wkljenwert

Why is it significant:

;lkwnerlv;knwer

Summary:

3 – Next Project

Lkwenrvlknwerv

Why is it significant:

We;rkjtng;wkljenwert

Why is it significant:

;lkwnerlv;knwer

Summary:

Authors Disclaimer

I declare that the work contained in this thesis was carried out in accordance with the regulations of the University of Gloucestershire and is original except where indicated by specific references in the text. No part of this thesis has been submitted as part of any other academic award. The thesis has not been presented to any other educational institution in the United Kingdom or overseas.

Any views expressed in the thesis are those of the author and in no way represent the views of the University.

I confirm that there has been no use of an editor at any point for the construction of this thesis.

Signed:



Date: 19 April 2025

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Abbreviations / Glossary Terms

Change this into two columns when ready to submit.

Term	Description
Atomograd	Atomic City (Russian for 'Atomic City').
ChNPP	Chernobyl Nuclear Power Plant.
CoM	Council of Ministers.
GammaNet	Wireless Sensor Network for Monitoring levels of Radiation and Environmental Factors in a given research location.
IoT	Internet Of Things.
NSC	New Safe Confinement.
UK	United Kingdom of Great Britain and Northern Ireland.
USSR	Union of Soviet Socialist Republic (Russia).
WSN	Wireless Sensor Network

Chapter 1 - Introduction:

1.1: Background

Ewr;fjbqe;rjfbn

1.2: Research Problem and Motivation

1.2.1: The Research Problem

K;djfbv;kjfbv

1.2.2: The Research Motivation

1.2.2.1: Personal Academic Motivation

My motivation for pursuing this research stems from a long-standing personal and academic interest in the Chernobyl Nuclear Power Plant (ChNPP) disaster of 1986. Since 2012, I have been deeply engaged with the subject, initially through documentaries and online content, including YouTube videos and news archives. During my undergraduate studies at the University of Gloucestershire, I expanded this interest by exploring a wide range of academic materials available in the university library, including books and documentaries detailing the incident and its aftermath.

A particularly formative moment in my academic journey was discovering the live webcam feed documenting the construction of the New Safe Confinement (NSC) structure—designed to contain the original sarcophagus built shortly after the explosion. Observing this engineering feat in real time further inspired me to investigate the disaster and its ongoing legacy more deeply.

This passion matured during my Master's degree studies, where I focused my thesis on investigating the operational limits of microcontrollers exposed to gamma radiation. This project allowed me to explore the intersection of electronics and radiological science, assessing how long embedded systems could function under irradiated conditions—an experience that directly shaped my vision for PhD research.

Initially, my PhD concept revolved around designing a robotic system capable of withstanding extreme radiation levels, with the ultimate goal of assisting in the clearance of the “Elephant’s Foot” — a highly radioactive mass located beneath Reactor No. 4. This corium formation, a mixture of nuclear fuel, control rods, sand, boron, and structural materials, formed during the reactor’s meltdown and flowed through the plant’s infrastructure. Although the project has since evolved in scope, this early idea reflects my commitment to contributing to the long-term scientific understanding and technological solutions related to the Chernobyl disaster.

1.2.2.2: Contextual Importance

Wdfiknv

1.2.2.3: Real World Relevance:

W;lefkjv

1.2.2.4: Research Gaps

Er;’Inw;erlijkvn

1.3: Research Questions or Hypothesis

Primary Question:

To what extent can IoT-enabled Wireless Sensor Networks (WSNs) be used to assess and monitor legacy radiological contamination in the UK resulting from the Chernobyl disaster?

Secondary / Supporting Research Question(s)

1.3.1: Environmental & Radiological Focus

1: What are the current levels of residual radiological contamination in areas of the UK affected by fallout from the Chernobyl disaster?

2: How accurately can historical meteorological and cloud movement data from 1986 be used to identify regions in the UK most affected by Chernobyl-related radionuclide deposition?

3: What are the long-term environmental impacts of the Chernobyl disaster on UK soil and ecosystems nearly four decades later?

1.3.2: Technical and System Design Focus

1: How can Internet of Things (IoT) technologies be effectively integrated into a Wireless Sensor Network (WSN) to support autonomous environmental monitoring?

2: What are the benefits and limitations of using a star network topology in the design of a scalable, field-deployable WSN for radiological monitoring?

3: How effective are 433 MHz RF communication modules in supporting long-range, low-power data transmission for real-time environmental sensing in remote UK regions?

1.3.3: System Evaluation & Deployment

Qw';fljq;ljkq;lksjdnv;aklsjdnv

1.3.4: Interdisciplinary Integration

Qe;jvn;qlejfbv

1.4: Aims and Objectives

1.4.1: Aims:

1: To evaluate the environmental and public health implications of Chernobyl-related radiation contamination in the UK by correlating radiological measurements with historical and current public health data (e.g., cancer rates, environmental toxicity reports).

2: To contribute to the scientific community's understanding of legacy contamination and its long-term persistence by providing real-time, high-resolution data that could inform environmental cleanup efforts, policymaking, and future disaster preparedness strategies.

3: To design, develop, and deploy an IoT-enabled Wireless Sensor Network (GammaNet) for monitoring and assessing the current environmental levels of

radiological contamination in areas of the United Kingdom affected by the Chernobyl Nuclear Power Plant disaster.

1.4.2: Objectives:

Qe;rlvkjqe;kjv ;qekjvn

1.5: Scope and Limitations

Qe;rlknqw;elrkjnv

1.6: Overview of this Thesis and Structure

1.6.1: Thesis Overview

1.6.2: Thesis Structure:

The thesis is structured as follows:

Chapter 1 – Introduction: Provides background to the research, outlines the problem, and presents the research questions and objectives.

Chapter 2 – Literature Review: Reviews existing literature on [your key areas, e.g., wireless sensor networks, environmental data systems, and network architectures], identifying gaps that this study seeks to address.

Chapter 3 – The Chernobyl Incident: A thorough investigation surrounding the run up to and including the incident and the aftermath and the events after.

Chapter 4 – Methodology: Describes the research design, methods of data collection, and analysis techniques used in the study.

Chapter 5 – System Design and Implementation: Details the hardware and software design, including ESP32 configurations, network models, and field deployment strategy.

Chapter 6 – Results and Analysis: Presents the findings of the study, including performance metrics, field data, and system evaluation.

Chapter 7 – Discussion: Interprets the results in relation to the theoretical framework and research questions, discussing implications and limitations.

Chapter 8 – Conclusion and Future Work: Summarizes the study's contributions, reflects on the research journey, and suggests directions for future research.

Chapter 9 – Evaluation & Reflection: ;lekrng;qlknerg;

Chapter 2 - Literature Review

Introduction

This chapter presents a comprehensive review of the existing literature relevant to the Chernobyl Nuclear Power Plant disaster and its long-term radiological impact, particularly within the United Kingdom. It also explores the technological landscape surrounding environmental monitoring through the integration of Wireless Sensor Networks (WSNs), Internet of Things (IoT) technologies, and radiation detection systems.

The chapter begins by contextualizing the Chernobyl incident, offering a detailed account of the events that unfolded on April 26th, 1986, and examining the immediate and long-term environmental consequences. Emphasis is placed on the dispersion of radioactive contaminants across Europe and the subsequent deposition over various regions of the UK, with a focus on how rainfall and meteorological conditions contributed to the uneven distribution of fallout. A review of the scientific literature on radiological contamination highlights the environmental persistence of radionuclides such as Cesium-137 and Strontium-90, including their effects on soil quality, agricultural productivity, water systems, and ecological health. Studies assessing the social, economic, and public health impacts of this contamination within the UK are also examined.

Following this, the chapter delves into the application of Wireless Sensor Networks (WSNs) and IoT technologies in environmental monitoring. It explores their technical characteristics, deployment challenges, and communication protocols — including the use of long-range, low-power frequencies such as 433MHz. The advantages of Star network topologies, scalability, and real-time data transmission are discussed, particularly in the context of monitoring large or remote geographic areas. Radiation sensor technology is also critically reviewed, evaluating various sensor types suitable for field deployment, such as Geiger-Müller tubes and solid-state detectors. Considerations for sensor accuracy, calibration, durability, and suitability for use in outdoor environments are explored.

Additionally, this chapter assesses existing national and international monitoring systems, such as the UK's Radioactive Incident Monitoring Network (RIMNET), identifying strengths, limitations, and gaps in coverage and responsiveness. Lessons learned from these systems inform the design goals of GammaNet, the wireless sensor network proposed in this study.

The review concludes by identifying key gaps in the existing literature, particularly the limited integration of modern IoT solutions into radiological monitoring infrastructures and the lack of localized, real-time environmental data collection in areas historically affected by Chernobyl fallout. These gaps provide a strong rationale for the current research and the development of an innovative, IoT-based monitoring system capable of supporting environmental assessments in the UK.

2.1 Review of Relevant and Previous Work

Everyone today knows the name 'Chernobyl' and its association with one of the world's worst nuclear disasters (second being Fukushima in 2011) and most recently portrayed in a 5-part mini-series produced by HBO and Sky UK^[1]. The incident which occurred on April 26th, 1986, is classed as a Level 7 (Major Incident) on International Nuclear and Radiological Events Scale (INES)^[2]. Originally developed in 1990 and updated in 2008, Chernobyl is widely considered the worse disaster in terms of human cost, radiation release, and long-term environmental damage. It was largely preventable and catastrophic in its execution and aftermath.

Fukushima^[4], while still extremely serious, was less severe in terms of human health impact and radiation exposure and was triggered by an unprecedented natural disaster.

2.1.1: The Chernobyl Incident

A comprehensive examination of the Chernobyl Incident is presented in Chapter 3, which was a particularly anticipated component of this research. In the following section.

2.1.2: Radiological Dispersal and Environmental Contamination across Europe.

2.1.2a: 01:23:40am

There were two explosions at the Chernobyl Nuclear Power Plant (ChNPP) on the night of April 26, 1986, each distinct in nature and consequence. The first explosion, occurring at approximately 01:23:44, was caused by a catastrophic steam buildup inside the reactor vessel. As the nuclear reaction spiralled out of control, the cooling water rapidly converted into steam due to the extreme heat. The enormous pressure was more than the system could withstand. A tremendous roar was heard in the control room, and the ground trembled violently, resembling the onset of an earthquake—these were the immediate physical effects of the steam explosion. Just moments earlier, at 01:23:40, Alexander Akimov, the shift supervisor, had pressed the AZ-5 emergency shutdown button, initiating the insertion of control rods into the reactor core. Ironically, due to a flawed control rod design involving graphite tips, this action triggered a sudden and massive spike in reactivity, leading directly to the first explosion.

The blast dislodged the Upper Biological Shield—nicknamed “Elena”—a massive steel and concrete disk that served as the reactor's protective lid. While its weight is subject to some debate, the International Atomic Energy Agency (IAEA) reported it to be 1,000 tonnes^[5], whereas author Serhii Plokhyl^[6] has cited an alternate (and likely erroneous) figure of 200 kg^[6]. The explosion sheared the pressure tubes at their welds and hurled the biological shield upward into the roof of the reactor building. Under the force of gravity, it crashed back down askew, landing on its side like a poorly flipped coin. This left the reactor exposed to the environment, allowing a torrent of radioactive material to begin spewing into the atmosphere almost immediately.

2.1.2b: The Second Explosion 01:23:44

01:23:44, operators in the control room felt a second but much more powerful blast, the floor and walls shook violently, those in the control room had no idea what had just happened. A Mixture of Hydrogen and Oxygen most likely ignited by the super red hot graphite, this second intense explosion ejected the highly radioactive

graphite blocks (the Moderator Core) as well as 7 tonnes of reactor fuel, damaged control rods zirconium channels were pulverised in to tiny fragments and sucked high up into the atmosphere forming a mixture of gasses and aerosols including: Iodine-131, neptunium 239, cesium-137, strontium-90 and plutonium-239, these are the most dangerous substances known to man, outside of a sealed nuclear reactor^[7].

2.1.3: The Chernobyl Cloud covering Europe and Entering the United Kingdom

Immediately after the two explosions at the Chernobyl Nuclear Power Plant on the 26th April 1986, a variety of radionuclides were forced high up into the atmosphere by the immense heat, this continued for a period of 10 days^[8] until the fire was extinguished inside the reactor. Radioactivity was first detected outside of the Soviet Union at a Nuclear Power Station in Sweden where monitored workers were noted to be contaminated^[9]. The radioactive cloud was tracked from Chernobyl as winds blew it in a north-westerly direction over the Soviet Union and into Europe, its movements were responsible for deposition of radionuclides in countries as far as Scandinavia, The Netherlands as well as Great Britain, the severity of these depositions depended on the height of the cloud/plume, the wind speed, the direction, the terrain the cloud went over as well as the amount of rainfall in those areas.

2.1.3.1: First Detections outside of the USSR:

Cliff Robinson, a radiation control officer raised the alarm at around 07:00 am on April 28th 1986 at the Forsmark Nuclear Power Plant near Uppsala in Sweden, 1257km away from incident at Chernobyl. The contamination was discovered when Cliff passed from one zone at the power plant to another via a Radiation Monitoring Point (RMP)^{[10] [11]}. Swedish diplomats contacted various Nuclear Agencies (Poland, East Germany the Soviet Union^[12], all of which denied any release of radionuclides into the atmosphere, however, pressure was put on the soviet ministers, and they could no longer conceal the facts.

Ignorant members of the Politburo (Principal policymaking committee in the Soviet Union) considered late in the morning of the 28th April 1986 whether to make a public announcement, they believed the contamination was limited to the territory of the

Soviet Union and that it was entirely a Soviet matter. Vladimir Dolgikh (See: Figure 11-1) stated that the radioactive contamination was only 60km in diameter, how ignorant and naive could he possibly be.

Soviet Media broke its silence on Monday the 28th April 1986, three days after the incident and more than 12 hours after the radioactive cloud had passed over Sweden. Actions/Events and Situations that occurred in the Soviet Union were normally kept a secret a typical soviet tradition^[1].

Most Countries in Europe experienced depositions of radionuclides mainly ¹³⁷Cs (Caesium-137) and ¹³⁴Cs (Caesium-134) as the plume passed over their respective countries^[9], the deposits varied due to varying factors as outlined above (2.1.3.1).

2.1.3.1: Cloud breaches the Borders of the United Kingdom

Winds from the South East carried the radioactive cloud further north west, the cloud broke the borders of the united Kingdom on the 1st May 1986 at around 23:00 according to cloud mapping data in 2005 by the former agency Institut de radioprotection et de sûreté nucléaire (IRSN)^[14] produced a simulation of path the radioactive cloud immediately from the point of explosion through to late in the evening of the 9th May 1986, a video has been published to You Tube^[15] (See: **Figure 11-2**).

2.2: Assessing and Evaluating the Met Office Data and IRSN Cloud Mapping Data from 1st May 1986 – 7th May 1986

A Freedom of Information Act (FOIA) Request was sent to the UK Met Office via their web site, requesting weather data for the period of the 1st May 1986 – 10th May 1986, a response was quickly received from them via email with a link to their archived data sets for the period that I was requesting (See: **Figure 11-3**) and (See: **Figure 11-4**).

2.2.1: 1st May 1986 21:00 – 2nd May 1986 09:00

2.2.1a: IRSN Chernobyl Cloud Data:

Late into the evening of the 1st May 1986 at around 22:30pm cloud data^[15] (See Cloud Data: **Figure 11-5**) shows the cloud from Chernobyl entered the United Kingdom on the southeast coast touching East Sussex and Kent.

2.2.1b: Met Office Weather Data:

According to the Daily Weather Summary from the Meteorological Office between 21:00 on the 1st May 1986 a 09:00 on the 2nd May 1986, there was no rain fall on the Southeast Coast (See: **Figure 11-6**).

2.2.2: 2nd May 1986 0900 – 2nd May 1986 21:00

2.2.2a: IRSN Chernobyl Cloud Data

By 09:00 on the 2nd May 1986 the cloud had moved further north and to the west of the UK covering most of Devon and into Cornwall, moving up in to South and mid Wales and venturing further north of England up past Birmingham and in to North Yorkshire, (See Cloud Data: **Figure 11-7**).

2.2.2b: Met Office Weather Data

There was very little rain captured my Met Office Sensors, a small amount of rain 0mm – 0.4mm in the county of Hampshire (See Met Office Data: **Figure 11-8**).

2.2.3: 2nd May 1986 21:00 – 3rd May 1986 09:00

2.2.3a: IRSN Chernobyl Cloud Data:

By 21:00 on the 2nd May 1986 the cloud containing radionuclides from the Chernobyl Nuclear Power Plant incident 6 days ago had now breached the south east borders of the Republic of Ireland, Wales was completely covered, the Isle of Man was covered as well as most of the North Central and North Eastern England the whole of the east coast of Scotland up to the Grampian areas (See Cloud Data: **Figure 11-10**).

2.2.3b: Met Office Weather Data

Met Office Data from the period between 21:00 on the 2nd May 1986 and 09:00 on the 3rd May 1986 (12 Hour Period) rain measurement sensors captured rain fall data

from trace amounts (0mm-0.4mm) in Southern and South West England, whereas in North Wales saw trace amounts to 1mm of rain, Isle of Man saw 6mm (Douglas area) and 5mm (Ramsey/Cranstal area), Cumbria saw between 3mm-11mm of rain, moving north to Scotland, measurements of 2mm of rain in the Glasgow area and 1mm near to Edinburgh (See Met Office Data: **Figure 11-10**).

2.2.4: 3rd May 1986 09:00 to 3rd May 1986 21:00

2.2.4a: IRSN Chernobyl Cloud Data

The cloud from the incident had moved farther north over the United Kingdom, a sort of a convex arc going from the South East (Kent Area) moving North West over Birmingham and to the top of North Wales (Holyhead area) moving across the Irish Sea into middle parts of Ireland, covering the Isle of Man and Cumbria, north England and over parts of Southern and Western Scotland (See Cloud Mapping Data: **Figure 11-11**).

2.2.4b: Met Office Weather Data:

Between the hours of 09:00 – 21:00 on the 3rd May 1986 weather data from the Met Office shows that there was rain between those times from trace amounts (0mm – 0.4mm) in central and Eastern areas of England with 3mm of rain in Eastern Kent, up to 1mm in areas around Suffolk an average of 1mm-5mm in Wales, later in the day up until 21:00 there was a lot of areas that experienced rain fall with the most over North Western areas of England and especially the Isle of Man seeing the most at 15mm, western Cumbria saw 9mm of rain areas in Northern Ireland saw 11mm of rain. (See: **Figure 11-12**).

2.2.5: 3rd May 1986 21:00 to 4th May 1986 09:00

2.2.5a: IRSN Chernobyl Cloud Data

IRSN Cloud data by 21:00 on the 3rd May 1986 showed that the cloud had moved farther North, Scotland was completely covered apart from the Scottish highland areas around Inverness, all of Northern Ireland was covered in the cloud, Southern and Central England and all of Wales was now clear of the cloud (See IRSN Cloud Mapping:).

2.2.5b: Met Office Weather Data:

Rainfall continued through the night of the 3rd May 1986 into the 4th of May, Met office data shows the period ending at 09:00 on the 3rd May 1986 there was rainfall on the West and Central parts of the United Kingdom

2.2.6: 4th May 1986 09:00 to 4th May 1986 21:00

2.2.6a: IRSN Chernobyl Cloud Data

2.2.6b: Met Office Weather Data:

2.2.7: 5th May 1986 09:00 – 5th May 1986 21:00

2.2.7a: IRSN Chernobyl Cloud Data

2.2.7b: Met Office Weather Data:

2.2.8: 5th May 1986 05:15am

2.2.8a: IRSN Chernobyl Cloud Data

At around 05:15 on the 5th May 1986, the cloud had cleared the borders of the UK, the winds had moved the cloud back over the North Sea and into The Netherlands and Sweden.

2.2.9b: Met Office Weather Data:

2.2.9: 7th May 1986 at 10:40am

2.2.9a: IRSN Chernobyl Cloud Data

IRSN displayed cloud data showing that small arm from the radioactive cloud moved back over towards the Shetland Islands, by 13:30 there was an increased level of radiation (Caesium-137) in the cloud over Shetland, by 18:35 the cloud had moved away from the Shetland Islands and gone further north towards the Faroe Islands, cloud mapping data ended on the 9th May 1986 at around 19:30.

2.2.9b: Met Office Weather Data:

2.3: Why is Rain and Weather data so important in this situation

Why is this rain information so significant for that time, with the cloud over most parts of the United Kingdom, rain will pull those radionuclides from the clouds and dispose of them on the ground, these radionuclide consisted mainly of ^{137}Cs (Caesium 137), ^{134}Cs (Caesium 134), ^{131}I (Iodine-131), ^{132}Te (Tellurium-132), ^{90}Sr (Strontium-90), ^{103}Ru (Ruthenium-103) & ^{106}Ru (Ruthenium-106), ^{95}Zr (Zirconium-95) and finally Plutonium isotopes ^{239}Pu and ^{240}Pu ^[7].

All the above isotopes have different half-lives meaning that once that the first half-life is finished it still has that amount to go before its completely gone or decayed into a more other or more stable element. So ^{137}Cs (Caesium-137) has a half-life of 30 years, we are just about on the 40th anniversary so we are 10 years into the second half-life stage of this isotope being present in the UK environment, ^{137}Cs (Caesium-137) will be present in our environment until the middle of 2046. ^{90}Sr (Strontium-90) has a half-life of 28.8 years, In January 2011 the first phase of its half-life had completed. We are now into the second half life phase of that isotope being present in the environment, so traces of ^{90}Sr (Strontium-90) will still be present in the UK environment until roughly the end of 2039.

Other Radionuclides that were dispersed over the UK during that period would have had shorter half-lives meaning that they would have decayed into safer elements along time ago, such as ^{131}I (Iodine-131) has a half live of 8 days meaning that 16 days after the explosion at Chernobyl it would have decayed to a more stable element, ^{134}Cs (Caesium-134) has a half-life of 2 years that means it will have decayed into a stable element by 1990, the Ruthenium isotopes ^{103}Ru and ^{106}Ru will have decayed within 76 days to 2 years respectively, the ^{95}Zr (Zirconium-95) will have decayed relatively quickly with only having a half-life of 64 days, meaning it will have decayed to a stable element within 130 days.

2.3.1: The UK's exposure to Fallout; Metrological Models, Contamination Levels and affected regions

2.4: Radiological Contamination and Environmental Impact of the UK

2.4.1: May 1986 Hydrological Review Data

During the First Week of May 1986, the UK Government advised the public not to drink rain water and stream water in Scotland and Northern England; runoff in these areas as the risk was heightened from the north-west drift of radioactive material following the major accident at the Chernobyl Nuclear Power Plant, in the USSR on April 26th [1]

Commented [MR1]: Natural Environment Research Council. (1988). Hydrological Data UK. Oxon UK: Institute of Hydrology. p.21.

2.4.2: Long-term effects of radionuclides such as Cesium-137 and Strontium-90.

2.4.3.1: What is Cesium-137

2.4.3.1a: How did Cesium-137 Affect the UK Environment

2.4.3.1b: How does Cesium-137 Impact on Soil

2.4.3.1c: How does cesium-137 Impact on Water

2.4.3.1d: How does Cesium-137 from Chernobyl Impact on Agriculture

2.4.3.1.e: How does Cesium-137 Impact on Ecosystem

2.4.4: What is Strontium-90

2.4.4.1a: How does Strontium-90 Impact on Soil

2.4.4.1b: How does Strontium-90 Impact on Water

2.4.4.1c: How does Strontium-90 Impact on Agriculture

2.4.4.1d: How does Strontium-90 Impact on Ecosystem

2.4.5: Case studies or reports on contamination in the UK post-1986.

Commented [MR2]: Paper: Cesium-137: Radio-Chemistry, Fate, and Transport, Remediation, and Future Concerns - Ashraf, Muhammad Aqeel

Commented [MR3]: Paper: Post-Chernobyl surveys of radiocaesium in soil, vegetation, wildlife and fungi in Great Britain - Chaplow, J. S

Commented [MR4]: Paper - The use of 137Cs to establish longer-term soil erosion rates on footpaths in the UK - <https://pubmed.ncbi.nlm.nih.gov/20627401/>

Commented [MR5R4]: Printed

2.5. Wireless Sensor Networks (WSNs) for Environmental Monitoring

2.5.1 Overview of Wireless Sensor Networks and their application in Environmental Science

2.5.2: WSN Design Considerations

2.5.2.1 Design Considerations

2.5.2.2: Network Topology

2.5.2.3: Power Consumptions

2.5.2.4: Data Transmission Protocols

2.5.3: Challenges in Deploying WSN

2.5.3.1: Deploying WSN in Rural Areas

2.5.3.2: Deploying WSN Harsh Environments

2.6. Internet of Things (IoT) in Environmental Monitoring

- Integration of IoT technologies with WSNs for real-time data acquisition.
- Case studies of IoT-based radiation monitoring systems.
- Benefits and limitations of using IoT in low-resource or remote areas.

2.7. Communication Technologies for Remote Sensing

- Use of 433 MHz and other low-power long-range communication protocols (e.g., LoRa, Zigbee).
- Data reliability and latency in environmental sensor networks.
- Comparative analysis of communication protocols for WSNs.

2.8. Radiation Detection and Sensor Technology

- Types of radiation sensors suitable for environmental monitoring (e.g., Geiger-Müller tubes, scintillation detectors).
- Sensor calibration and accuracy in low-level contamination areas.
- Durability of sensors in outdoor and potentially radioactive conditions.

2.9. Previous and Existing Monitoring Systems

- Legacy and modern radiation monitoring networks in Europe and the UK (e.g., RIMNET).
- Evaluation of their effectiveness, data accessibility, and technological limitations.
- Lessons learned from past deployment of monitoring infrastructure.

2.10. Policy, Regulation, and Public Health

- Governmental response to Chernobyl in the UK.
- Regulations and safety limits on radioactive contamination in food and soil.
- Public health implications and long-term studies on affected populations.

2.11. Gaps in the Literature

- Where are the current blind spots in research or monitoring?
- Limitations in existing models of contamination spread or current monitoring infrastructures.
- Opportunities for innovation, such as systems like your proposed **GammaNet**.

2.12. Optional: Interdisciplinary Connections

- Ethics of nuclear disaster research and monitoring.
- Risk perception and communication to the public.
- The role of open-source hardware and citizen science in radiation monitoring.

2.3: Theoretical Foundations

Qervwerv

2.4: Identification of Research Gaps

Qervqe

2.5: Positioning your Work in the Existing Field

Cwqvqevqwev

2.6: Chapter Conclusion

Chapter 3 - The Chernobyl Incident – An In-depth Review

Introduction

In this chapter, I will provide an in-depth exploration of the Chernobyl Nuclear Power Plant (ChNPP) disaster, analysing the events and activities that led to the incident, the course of the disaster itself, and the immediate aftermath. I will investigate those responsible for the crisis, the remedies that were implemented in response, and the current state of Chernobyl today. Additionally, I will examine the far-reaching impact of the incident on the United Kingdom, including its effects on the environment, agriculture, and the people who lived in the affected areas. Some sources/references in this chapter can be quite dated due to the incident happening back in 1986 some 40 years ago now, there have been many studies in relation to the Incident, so I will do my best to use the most up to date sources ,however, I will have to use sources from back in the 80s.

3.1: Let's start at the beginning

The decision to construct the Chernobyl Nuclear Power Plant (ChNPP) came about in the mid 1960's (1966 to be exact) the Council of Ministers (CoM) of the Soviet Unions (Union of Soviet Socialist Republic(USSR)) the plan was to have 6 reactors at the power plant each producing 1,000 Mwe of electricity to the Soviet Union^[1]. After surveying 16 possible locations for the new power station in Kyiv, Vinnytsia and Zhytomyr regions, the ministers narrowed down the locations to just 2 options, near to the village of Ladyzhyn (Haysynsky District, south of the River Buh) (See: **Figure 11-14**) and the other option was near the village of Kopachi (Chernobylsky District) (See: **Figure 11-15**). The Ministers settled on the Kopachi region on the left side of the Pripyat river, 15km North of the town of Chernobyl (See: **Figure 11-16**).

3.1.1: Starting Construction of the Chernobyl Nuclear Power Plant

Viktor Bryukhanov (See: **Figure 11-17**) was named as Director in 1970 and was also the head of the directorate of the Chernobyl Nuclear Power Plant (ChNPP), Viktor was also the power station director up until the incident in 1986. Surveying began early in 1971 marking out locations on the site where everything was to be placed

(See: **Figure 11-18**). The first Cubic metre of concrete for reactor block 1 was poured on the 15th August 1972 during a ceremony where it marked the start of the construction of the main buildings (See: **Figure 11-19**) of the Chernobyl Nuclear Power Plant.

3.1.2: Building of Pripyat (An Atomograd - The Workers Town)

In February of 1970, construction began to build the town that would house the workers and their families (See: **Figure 11-20**), the town was located 3km northwest of the power plant construction site, the town was named Pripyat (An Atomograd) (See: **Figure 11-21**), it was estimated to house around 75-85 thousand people^[5].

During the construction of the power plant and before Pripyat was ready, construction workers stayed in dormitories or with local villagers, Workers rented apartments within a 20km radius of the construction site, like what is happening today with the construction of Hinkley Point C^[8] (New Nuclear Build).

3.2: Construction Delays and setbacks

As with any large-scale project there was delays and setbacks, the first reactor at the Chernobyl Nuclear Power Plant was due to be turned on in 1974, however the general contracting company “Kremenchukgesbud Trust” which is a construction department of the Ministry of the Energy and Electrification of the USSR had never built nuclear power stations before, the construction plans were not being followed, and construction and engineering levels were low with high turnover of construction staff and construction machinery is not being used^[5].

The switch on of reactor one was delayed until 1975 by the Ministry of Energy of the USSR, due to the interruption of the supply of necessary equipment, however, the First Secretary of the Central Committee of the of the Communist Party of Ukraine (CC CPU) Volodymyr Scherbatsky (See: **Figure 11-22**) had to personally complain to the Chairman of the Council of Ministers Oleksiy Kosygin (See: **Figure 11-23**), after this complaint, supplies started to come through more regularly, however, the start-up of the first reactor was postponed for a second time to 1977.

It was decided that there would be no third request to postpone the start-up of reactor one, so construction workers worked as increased speed, in some areas they worked around the clock to get sections finished. On Monday 26th September 1977 at around 8:20pm, turbogenerator No.2 of the first block was turned on, from that point on the first nuclear powered electricity in Ukraine started flowing from the plant to the Electricity grid.

3.2.1: The Reactors at Chernobyl Nuclear Power Plant

“The RBMK is safe enough to be installed on the Red Square”

– Anatolii Aleksandrov^[11]

3.2.1a: The RBMK 1000 High Power Channel Reactor

The RBMK-1000 (Reaktor Bolshoy Moshchnosty Kanalny) High Power Channel Reactor (See: **Figure 11-27**) was a creation between Anatolii Aleksandrov (See: **Figure 11-24**)^[12], who at the time was the President of the Soviet Academy of Sciences and Yefim Slavsky (See: **Figure 11-25**) who held the director's position of the Ministry of Medium Machine Building (A top Secret Ministry). The RBMK-1000's chief designer Nikolai Dollezhal (See: **Figure 11-26**), tested the reactor in Slavsky's ministry and it was deemed safe enough to be transferred to the Ministry of Energy and Electrification, which at the time had no experience with nuclear Energy. RBMK reactors were placed all over the European part of the Soviet Union (See: **Table 11-1**), 17 were successfully installed between 1973 at unit 1 at Leningrad and the last one being installed in 1990 at Smolensk, 4 were installed at the Chernobyl Nuclear Power Plant with 2 additional RBMKs planned for reactor buildings 5 and 6, however planning and development was halted after the incident in 1986.

With a capacity of 1000 MWe (Megawatts of Electricity, they out surpassed the Soviet competitor the VVER (Water-Water Energy Reactor) what had been produced since the early 1970s. By 1982 half of the country's' electricity supply had come from the RBMK reactor. The fourth reactor at Chernobyl had been started by Viktor Bryukhanov (See: **Figure 11-17**) in 1983.

In 1938 scientists Otto Hann (See: **Figure 11-28**) and Fritz Strassmann (See: **Figure 11-29**) in Germany discovered that when atoms of the heavy metal Uranium are bombarded with neutrons, their nuclei can be broken apart releasing energy, when the nuclei splits their neutrons fly away at high speed smashing into other atoms causing their atoms to split causing a unstoppable chain reaction. At 08:16am on the 6th August 1945 a fission weapon containing 64 kilograms of uranium detonated 580 metres above the Japanese city of Hiroshima, within a fraction of a second a city was levelled, some 78 thousand people died instantly or immediately afterwards were burned, vapourised or incinerated by the firestorm the followed the blast wave.

3.2.1b: The Atom

The atoms of different elements vary be weight, determined by the number of protons and neutrons in each nucleus. Each element has a unique number of protons which never changes, determining its atomic number and its position on the periodic table (See: **Figure 11-30**). The Periodic Table was created Dmitri Mendeleev (See: **Figure 11-31**) in 1869 when Dmitri started collecting properties of known elements and putting them into an ordered grid.

3.2.1c: Back to the RBMK-1000 Reactor

Back to the RBMK-1000 2nd Generation reactor, it is a water cooled graphite moderated reactor with individual fuel channels surrounded by graphite blocks (See: **Figure 11-32**), The reactor sits in a concrete vault measuring 25m deep and a diameter of 21m x 21m, the core of the reactor is made up of graphite blocks stacked on top of each other, similar to Lego bricks (See: **Figure 11-32**), each block measures 250mm x 250mm and vary in length from 200, 300mm, 500mm and 600mm (See: **Figure 11-32**) 600mm length being the most common^[20], The graphite blocks are arranged in a cylindrical configuration filling the vault, resulting in 1872 graphite columns measuring 8m tall, the total mass of the graphite blocks inside the reactor vault is roughly 1700 tonnes^[20]. 1661 of these channels are for Fuel and the other 211 channels are for the control rods (See: **Figure 11-37**). The control rods will become a fundamental factor in the catastrophe later on in this chapter.

3.3: Friday 25th April 1986 Pripyat

It was a warm sunny day almost like a warm summer day compared to early spring, residents of the town of Prip'yat (See: **Figure 11-38**) the 9th atomograd (Atom City) located 3km northwest from the Chernobyl Power Plant were getting ready for a long weekend leading up to the May Day bank holiday celebrations, engineering specialists were underway getting the cities new amusement park ready (See:) for the Labour Day celebrations on 1st May 1986, however, due to the incident this amusement park would never open^[23]. Families were also getting ready for the celebrations, home improvements, laundry, gardening.

“Hai bude atom robitnikom, a ne soldatom!”
– “Let the atom be a worker, not a soldier!”^[24]

Viktor Bryukhanov was always was under a conservable amount of stress from higher ups in the central committee and communist party he had left for work early from his apartment overlooking Kurchatov street in the town, just a short drive to the powerplant, Valentina (Viktors Wife) (See: **Figure 11-17**) had decided to take the afternoon off from working on the construction offices to spend time with her daughter and son-in-law.

Later that day and late into the evening at around 11pm, Leonid Toptunov (See: **Figure 11-39**) caught the workers bus that shuttled the midnight shift workers to and from the power plant from the bus station in the town of Prip'yat, he sat among his colleagues Boris Stolyarchuk (See: **Figure 11-40**) and Alexander Akimov (See: **Figure 11-41**) along with Alexander Yuvchenko (See: **Figure 11-42**)

3.#: Chapter Conclusion

Chapter 4 - Methodology

Introduction

In this chapter I will explain the smaller individual research project that I conducted building up to the bigger field research project when I deploy Sensor Nodes in locations around the United Kingdom.

4.1: Microcontroller (Arduino) Radiation Tolerance Investigation (MaRTI)

As previously mentioned, I originally intended to conduct this experiment for my Master's project in 2020. However, the onset of the COVID-19 pandemic led to a global lockdown, resulting in the closure of all non-essential scientific laboratories, including the Dalton Cumbrian Facility (DCF). With staff required to work from home, I was unable to carry out the experiment as planned. Consequently, I had to undertake a desk-based theoretical evaluation, relying on the analysis of previous and comparable experiments, a similar research project had been conducted at the National Aeronautics and Space Administration (NASA) Goddard Space flight Centre (GSFC) for their CubeSat programme^[1], which entailed irradiating a number of Arduino Microcontrollers with the ATMEL 328 8-bit microcontroller and the Raspberry Pi, the shielding used in this experiment was a mixture of Aluminium and Lead blend, and the source of the gamma radiation is unknown at this point.

This experiment aimed to determine the operational longevity of a microcontroller (Arduino) when exposed to a concentrated source of gamma radiation (Cobalt-60) at the Dalton Cumbrian Facility (DCF). The rationale for this investigation stems from the original plan for my Ph.D. field research project, which involved deploying a Wireless Sensor Network in the abandoned city of Pripyat, Ukraine, to collect radiation data over a 12-month period. As the sensor nodes would be placed in areas with high levels of radiation, the study also explored suitable shielding materials to protect the microcontroller and prolong its functional lifespan under such extreme conditions.

4.1.1 Research Design

The planning of this experiment had already been done during my Master individual project back in 2019-2020, The plan was to irradiate a number of Arduino microcontrollers with a concentrated source of gamma radiation, which meant I had to search the United Kingdom to find a laboratory or facilities that had such facilities, I started by enquiring with nuclear power stations, Hinkley Point B, Sellafield and others, and enquiring with the Office of Nuclear Regulations (ONR) to ask for their help, when it was identified that I needed to get in touch with the National Nuclear Laboratory (NNL) in Cumbria at the Dalton Cumbrian Facility (DCF) which is now apart of the Henri Royce Institute (HRI) in collaboration with The University of Manchester. I send back and forth email to Dr Ruth Edge to set up this experiment.

4.1.1a – Research Planning

Securing funding to carry out this experiment was of particular concern as I was only entitled to a very small amount of funding from my university which in the grand scheme of things is a pitiful amount, so I was advised by Ruth Edge to look to external funding and signposted to the National Nuclear Users Facility (NNUF) which provides grants and funding for experiments in the nuclear/radiation areas.

I applied for this grant with the support of my primary supervisor Professor Shujun Zhang and was awarded the grant for £2000-£5000, this allowed me access to the Gamma Irradiator (See: **Figure 11-43** & **Figure 11-46**) at the Dalton Cumbrian Facility (DCF).

4.1.1b: Research Rationale

The increasing reliance on microcontroller-based systems in environments exposed to varying levels of radiation—such as aerospace, nuclear, and medical applications—necessitates a deeper understanding of how such components perform under these conditions. While commercial microcontrollers like the Arduino platform are widely used due to their affordability and accessibility, their radiation tolerance remains underexplored, particularly in comparison to space-grade or hardened systems.

Current literature tends to focus on high-cost, specialized hardware designed for extreme environments, with limited attention given to the resilience of low-cost microcontrollers. This presents a critical gap, especially as space agencies, educational institutions, and small-scale researchers begin exploring more budget-conscious approaches to space and high-radiation experimentation.

The MaRTI project addresses this gap by systematically investigating the behaviour and fault tolerance of Arduino microcontrollers when exposed to radiation sources. Through controlled testing and analysis, this research aims to evaluate performance degradation, data corruption, and potential failure modes, thereby providing a clearer picture of their suitability for deployment in radiation-prone settings.

This study is significant in its potential to influence design decisions in cost-sensitive radiation environments, inform best practices for using COTS (Commercial Off-The-Shelf) electronics in research and industry, and contribute to the growing body of knowledge in radiation effects on embedded systems.

4.1.2: Instruments, Tools and Equipment

4.1.2a: The Equipment

For this investigation the most important pieces of equipment are as follows:

- 1) Arduino Uno Rev 3 (See: **Figure 11-47** & **Figure 11-48**)^[2 & 3],
- 2) 16x2 Liquid Crystal Display with built in I2C Controller (See:)^[4].
- 3) Foss Therapies Model 812 Self Contained Gamma Irradiator (See:)
- 4) Colbalt 60 (The Source of Gamma Radiation).
- 5) Various Shielding Materials
 - a) 10mm Thick Lead.
 - b) Code 3 Lead Flashing (1.32mm Thick – Midland Lead
 - c) PLA 10mm Thick
 - d) PLA+ 10mm Thick
 - e) PETG 10mm Thick
 - f) Steel 10mm Thick
 - g) Aluminium 10mm Thick
 - h) Concrete 10mm Thick.

4.1.2b: The Irradiation Dose Rates and Irradiation times:

Cycle	Dose Rate	Time	TID	cTID
1	10Gy/Min	5 Mins	50Gy	50Gy
2	20Gy/Min	5 Mins	100Gy	150Gy
3	30Gy/Min	5 Mins	150Gy	300Gy
4	40Gy/Min	5 Mins	200Gy	500Gy
5	50Gy/Min	5 Mins	250Gy	750Gy
6	60Gy/Min	5 Mins	300Gy	1050Gy
7	70Gy/Min	5 Mins	350Gy	1400Gy
8	80Gy/Min	5 Mins	400Gy	1800Gy
9	90Gy/Min	5 Mins	450Gy	2250Gy
10	100Gy/Min	5 Mins	500Gy	2750Gy

Table 4-1: Irradiation Dose Rates and Irradiation Times

TID = Total Ionising Dose.

cTID - Combined Total Ionising Dose.

4.1.2c: Set up of the Arduino and LCD Display

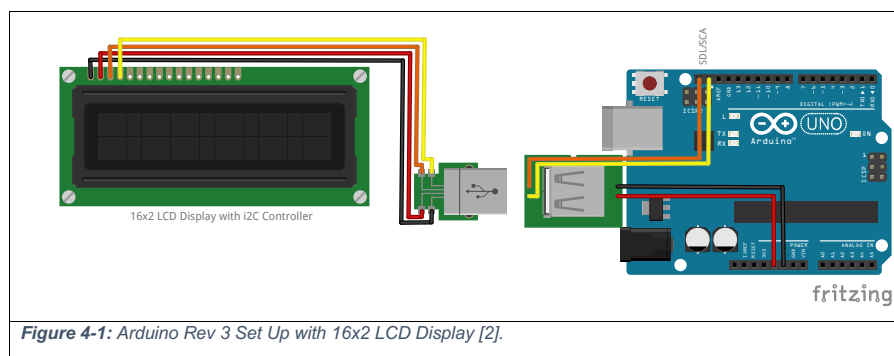


Figure 4-1: Arduino Rev 3 Set Up with 16x2 LCD Display [2].

The Software programme Fritzing¹ was used to lay out the components and then this images was exported in Scalable Vector Format (.svg) format and placed in this report and then also placed in the individual paper.

4.1.3: Data Collection

4.1.4: Data Analysis

4.1.5: Justification of Methods

4.1.6: Ethical & Safety Considerations.

4.2: Final Field Research Project – Phase One (Puriton)

Deploying the Sensor Nodes in the Village of Puriton

4.2.1 Research Design

4.2.2: Data Collection

4.2.3: Instruments, Tools and Equipment

4.2.4: Data Analysis

4.2.5: Justification of Methods

4.2.6: Ethical & Safety Considerations.

4.3: Final Field Research Project – Phase Two (Bridgwater)

Deploying the Sensor Nodes in and around the Town of Bridgwater

4.3.1 Research Design

4.3.2: Data Collection

4.3.3: Instruments, Tools and Equipment

4.3.4: Data Analysis

4.3.5: Justification of Methods

4.3.6: Ethical & Safety Considerations.

4.4: Final Field Research Project – Phase Three (Isle of Man)

Deploying the Sensor Nodes in and around the Isle of Man

4.4.1 Research Design

4.4.2: Data Collection

4.4.3: Instruments, Tools and Equipment

4.4.4: Data Analysis

4.4.5: Justification of Methods

4.4.6: Ethical & Safety Considerations.

4.4: Final Field Research Project – Phase Four (Cumbria)

Deploying the Sensor Nodes in and around the area of Cumbria

4.4.1 Research Design

4.4.2: Data Collection

4.4.3: Instruments, Tools and Equipment

4.4.4: Data Analysis

4.4.5: Justification of Methods

4.4.6: Ethical & Safety Considerations.

Chapter 5 - System Design / Implementation / Experimental Setup

Introduction

5.1: Architecture or Systems Descriptions (Hardware/Software)

5.2: Design decisions and Rationale

5.3: Technical Implementation

5.4: Field Work Setup

Chapter 6 - Results

Introduction

6.1: Results for the Microcontroller (Arduino) Radiation Tolerance Investigation

6.1.1: Presentation of Findings

6.1.2: Tables, Graphs Charts and Visuals

6.1.3: Raw Data (Summary only) Links to Raw Data in Appendix Section.

6.2: Results from:

6.2.1: Presentation of Findings

6.2.2: Tables, Graphs Charts and Visuals

6.2.3: Raw Data (Summary only) Links to Raw Data in Appendix Section.

6.3: Results from:

6.3.1: Presentation of Findings

6.3.2: Tables, Graphs Charts and Visuals

6.3.3: Raw Data (Summary only) Links to Raw Data in Appendix Section.

Chapter 7 - Discussion of Results

Introduction

7.1: Interpretation of Results

7.2: Comparison With Existing Work

7.3: Explanation of findings and Anomalies

7.4: Implications of the Results.

Chapter 8 - Conclusions

Introduction

8.1: Summary of Contributions

8.2: Main Findings

8.3: How your Work Answers the Research Questions

8.4: Limitations of Your Study

8.5: Suggestions for Future Research

Chapter 9 - Evaluation And Reflection

Introduction

I this chapter

9.1: Evaluation

;wejklrn;wljknerv

9.2: Reflection

Wer;jkvnw;erjklrv

9.3: Future Development

9.3.1: Future Development/Research of the GammaNet System.

Looking at the GammaNet system now in April 2025 (5 Months before first Deployment) and what I could add in future revisions of the system, I would look to apply miniature scintillator detectors to each sensor node, especially where the sensor node is close to the ground or add it to the sub surface probe so that I can detect Counts Per Second (CPM) but also detect the type of beta or gamma emissions as they pass through the tube using a Photomultiplier and Scintillator Crystals, as each emission will have a specific value of KeV (Kilo Electron Vaults) as the emission passes through the crystal it creates a light pulse and the photomultiplier detects this light and can determine its KeV value.

9.4: What's Next

Chapter 10 - References (By Chapter):

Chapter 1

Chapter 2

1. Chernobyl. (2019). 1:23:45. HBO & Sky UK, 6th May 2019.

Chapter 3

Chapter 4

Chapter 5

Chapter 6

Chapter 7

Chapter 8

Chapter 9

Chapter 11 - Appendix

Chapter 1

Chapter 2

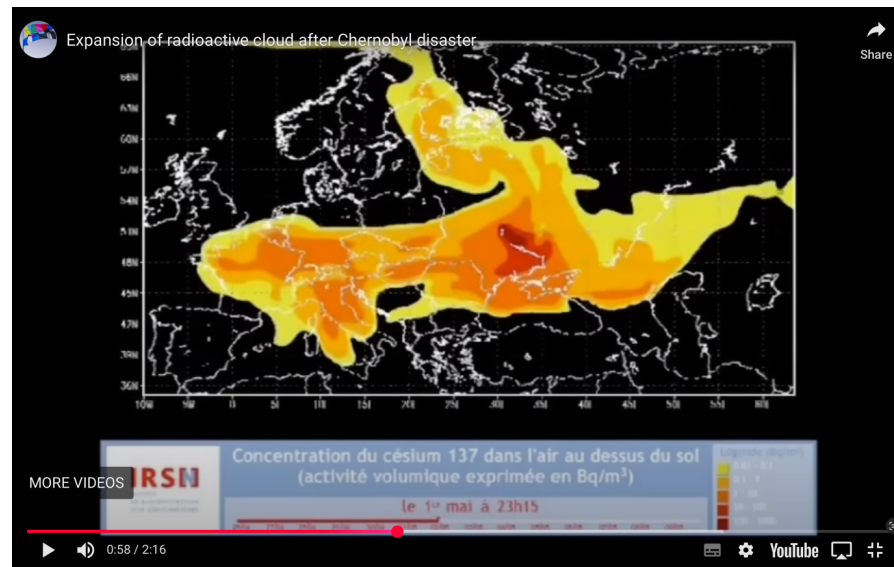


Figure 11-2: IRSN Chernobyl Cloud Mapping Data [15].

Email response from the UK Met Office

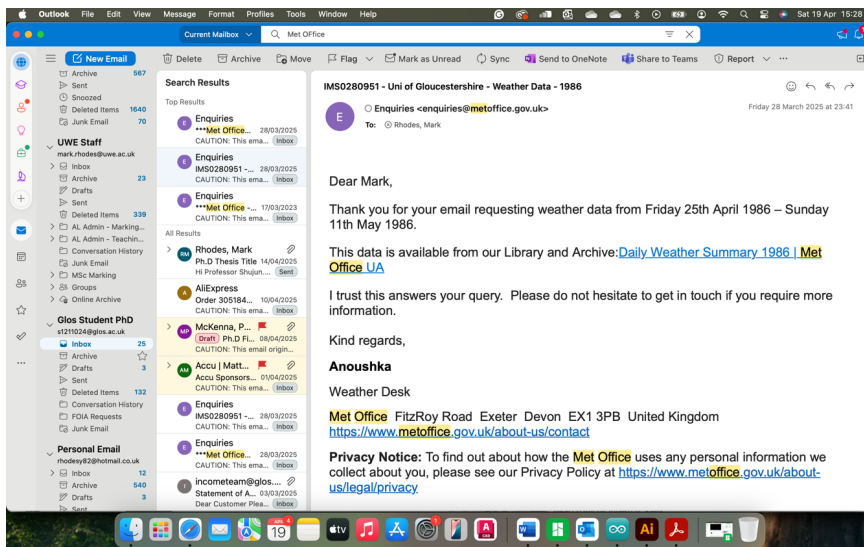


Figure 11-3: Email Response from Met Office [16]

Met Office Library and Meteorological Archive

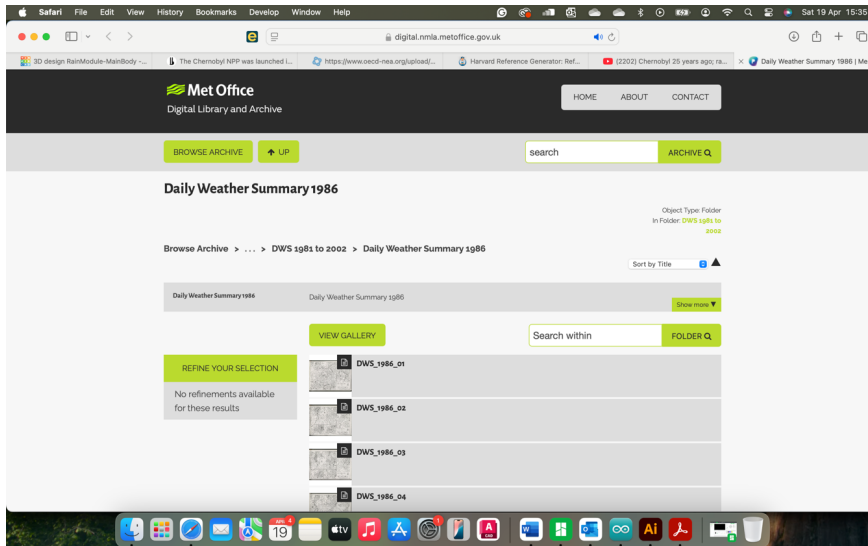


Figure 11-4: Met Office Digital Library and Archive [17].

IRS N & Met Office Cloud Mapping Data/Imaging

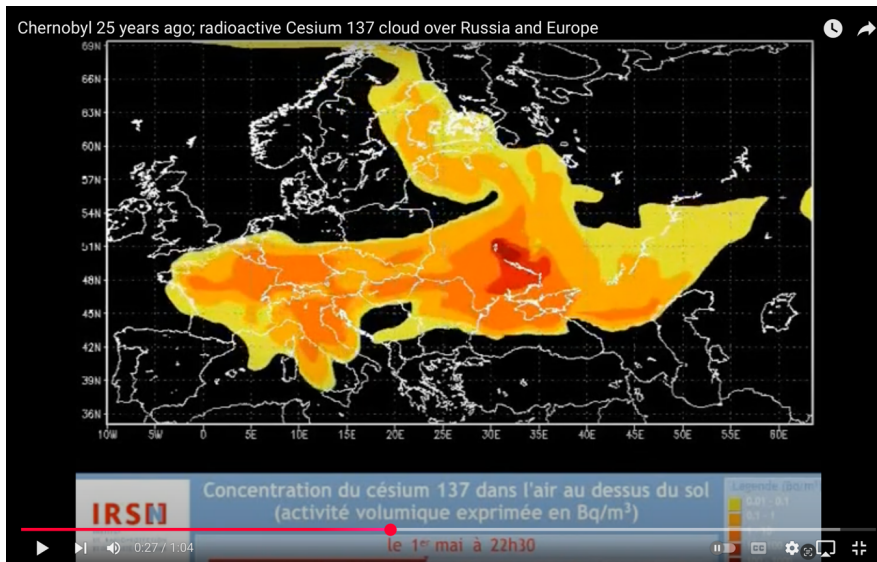


Figure 11-5: IRSN Chernobyl Cloud Mapping over UK on 1st May 1986 22:30 [15].

<https://www.youtube.com/watch?v=2ApKvCtHGDc>

1st May 1986 21:00 – 2nd May 1986 09:00 Weather & Cloud Mapping Data

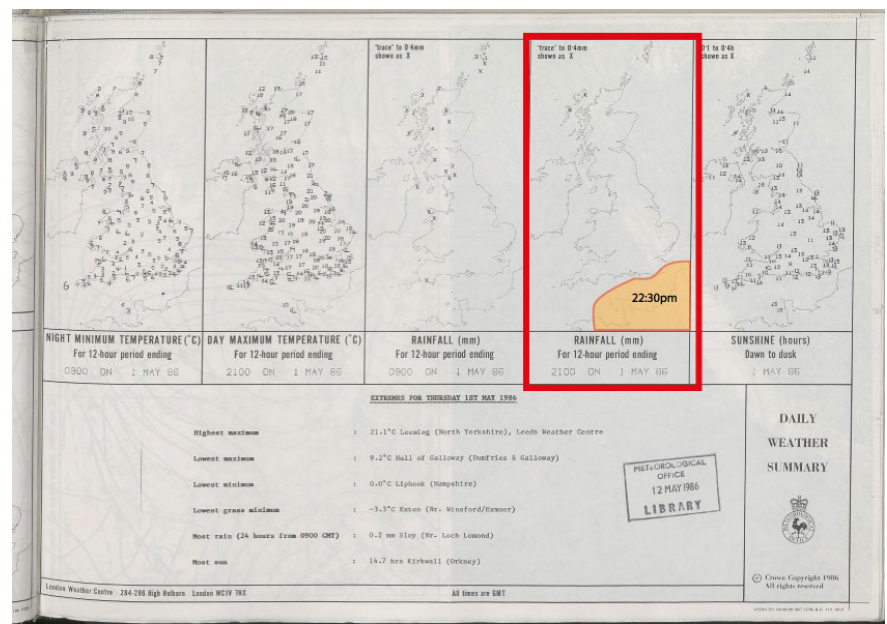


Figure 11-6: Met Office Weather Data and Chernobyl Cloud 1st May 1986 @ 22:30 []

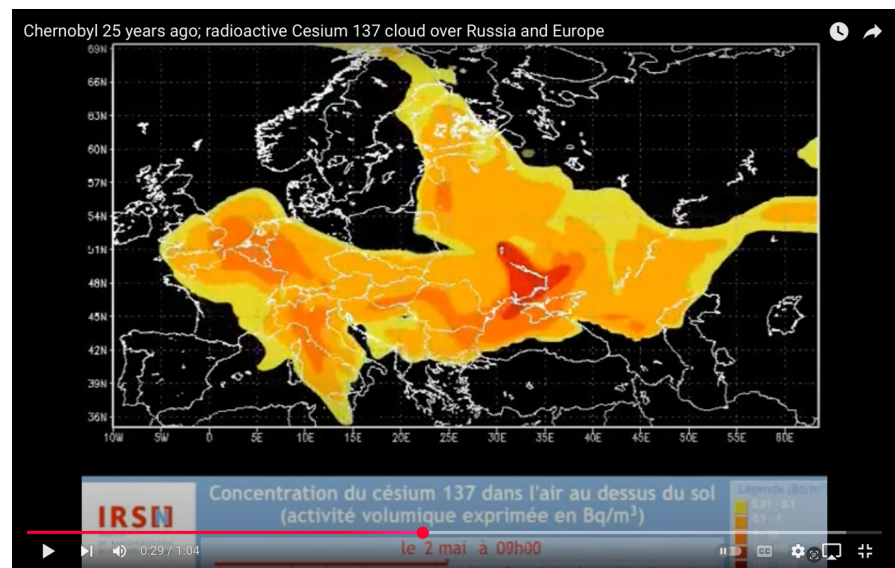


Figure 11-7: IRSN Chernobyl Cloud Mapping over UK on 2nd May 1986 09:00 [15].

2nd May 1986 21:00 – 3rd May 1986 09:00 Weather & Cloud Mapping Data

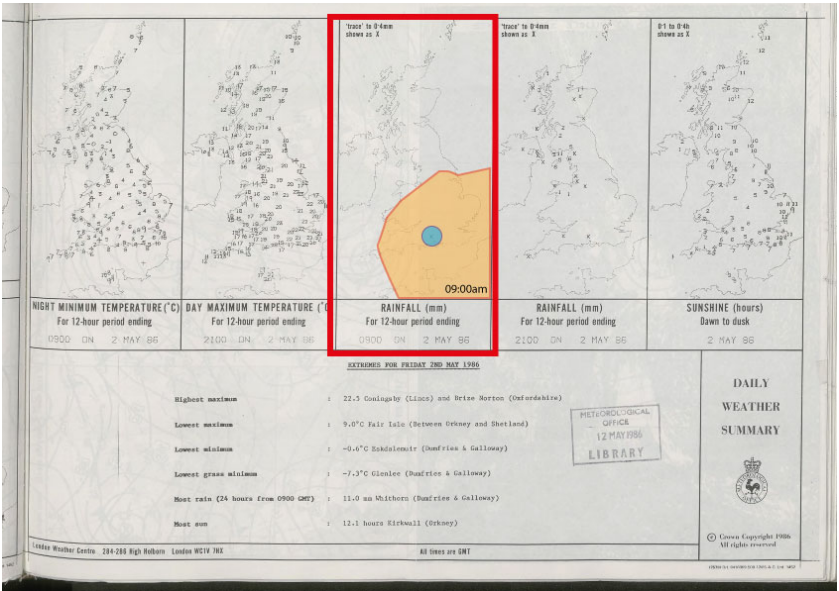


Figure 11-8: Met Office Weather Data and Chernobyl Cloud 2nd May 1986 @ 09:00 – 21:00 [7]

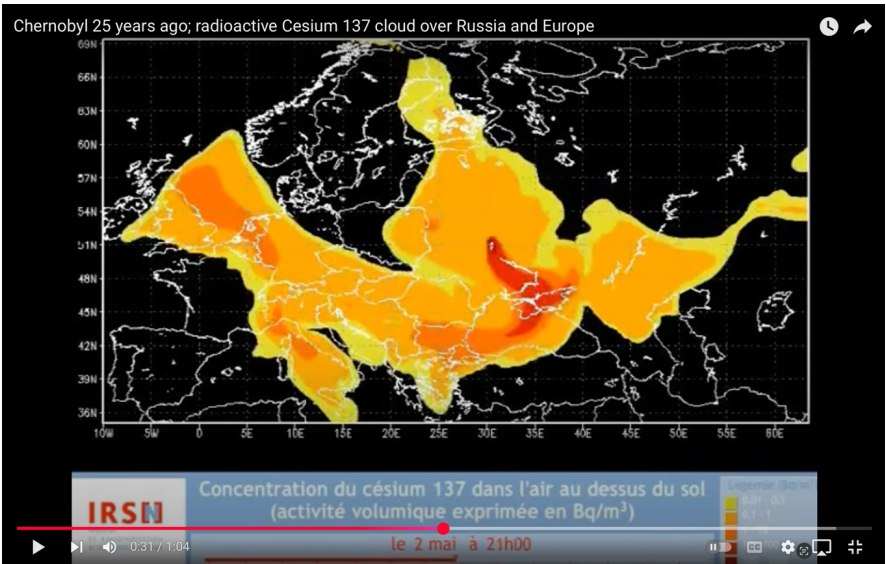


Figure 11-9: IRSN Chernobyl Cloud Mapping over UK on 2nd May 1986 21:00 [15].

3rd May 1986 21:00 – 4th May 1986 09:00 Weather & Cloud Mapping Data

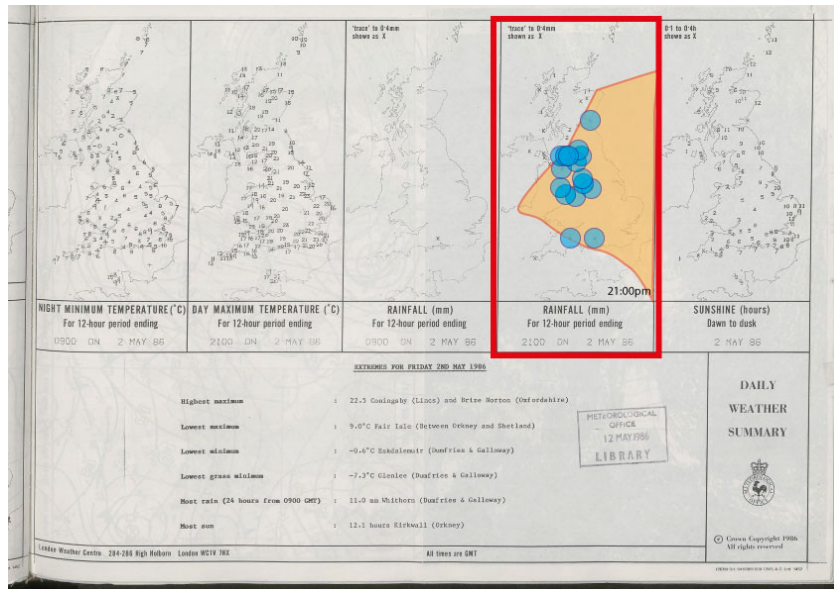


Figure 11-10: Met Office Weather Data and Chernobyl Cloud 2nd May 1986 @ 09:00 - 21:00 [15]

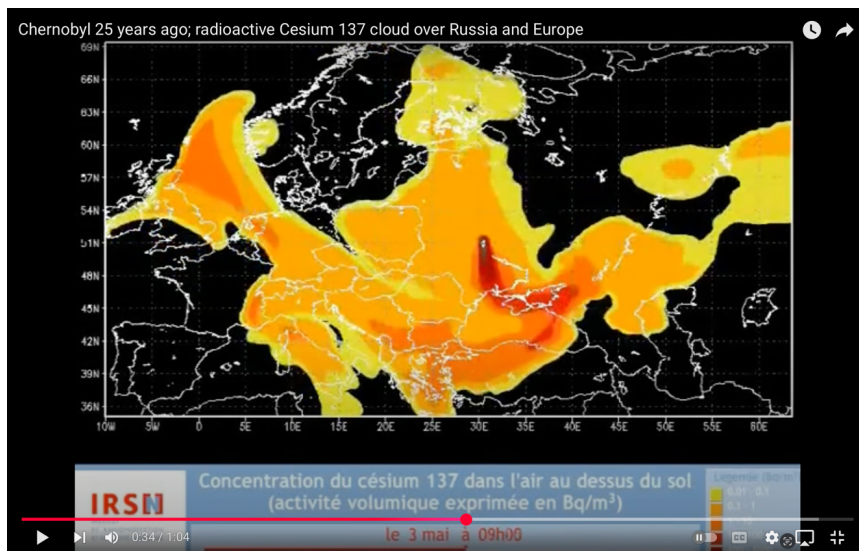


Figure 11-11: IRSN Chernobyl Cloud Mapping over UK on 3rd May 1986 09:00 [15].

4th May 1986 21:00 – 5th May 1986 09:00 Weather & Cloud Mapping Data

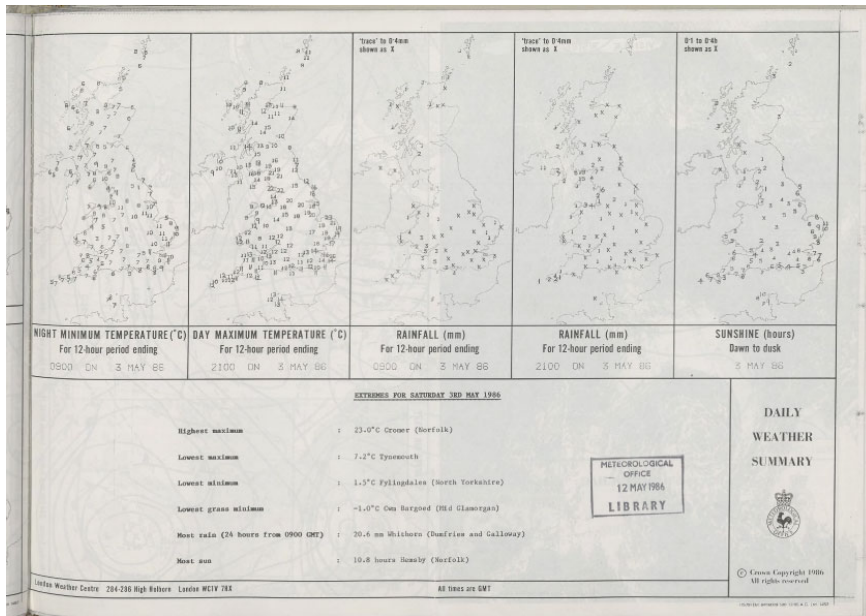


Figure 11-12: Met Office Weather Data and Chernobyl Cloud 3rd May 1986 @ 09:00 - 21:00 []

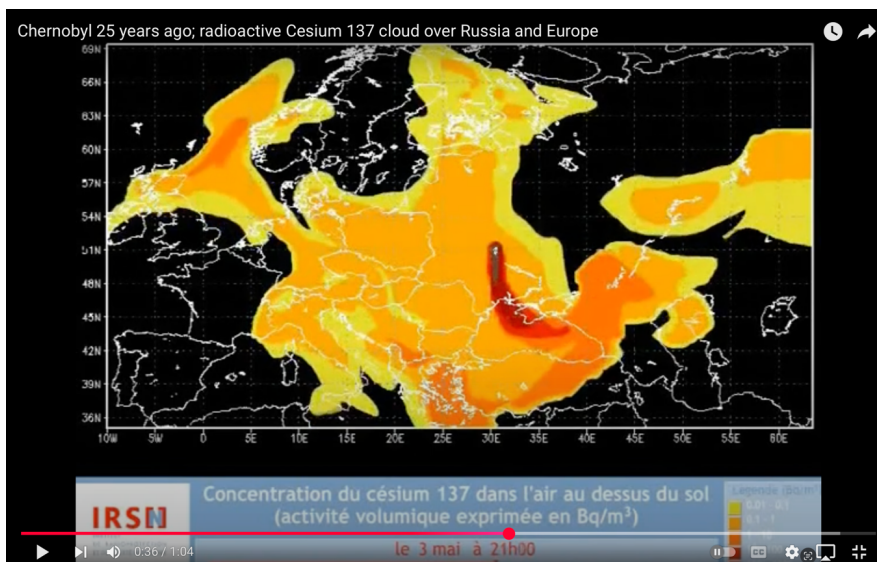


Figure 11-13: IRSN Chernobyl Cloud Mapping over UK on 3rd May 1986 21:00 [15].

5th May 1986 21:00 – 6th May 1986 09:00 Weather & Cloud Mapping Data

Cloud Data

Met Office Data

6th May 1986 21:00 – 7th May 1986 09:00 Weather & Cloud Mapping Data

Cloud Mapping Data

Met Office Data

7th May 1986 21:00 – 8th May 1986 09:00 Weather & Cloud Mapping Data

Cloud Mapping Data

Met Office Data

Chapter 3

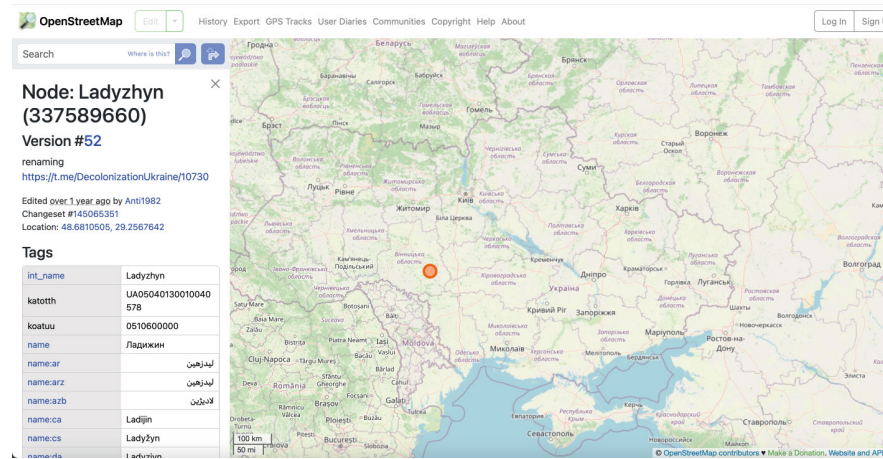


Figure 11-14: Ladyzhyn Possible Location of the Power Plant Construction in 1960's.

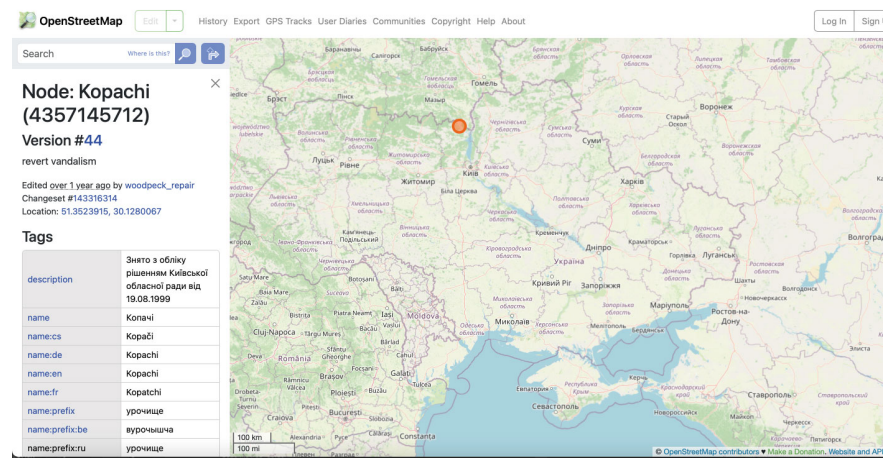


Figure 11-15: Kopachi Possible Location of the Power Plant Construction in 1960's

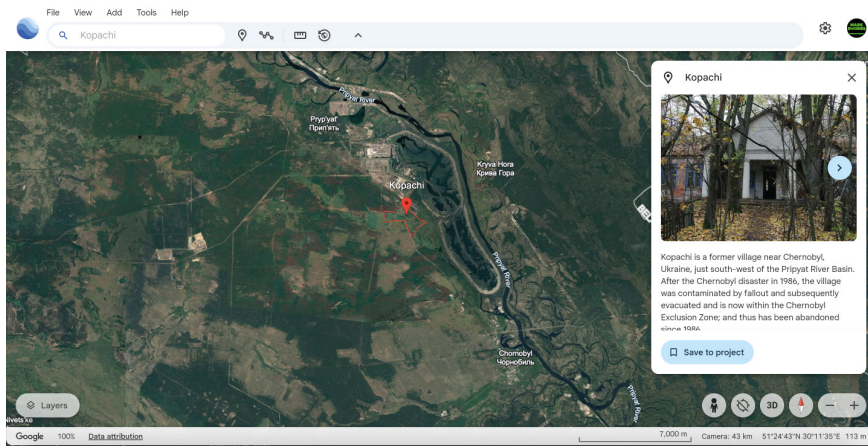


Figure 11-16: Google Earth Image of the Kopachi Region relative to Chernobyl Town.



Figure 11-17: Viktor Bryukhanov and his wife Valentina, 1980 or so [6].



Figure 11-18: Surveying Areas for the Chernobyl NPP, 25th May 1971. [5]



Figure 11-19: Pouring of the First Cubic Metres of Concrete at the ChNPP, 15th August 1971 [5]



Figure 11-20: Construction of residential buildings in the town of Pripjat (Atomograd), February 1971 [5]



Figure 11-21: Prip'yat Town Sign (Circa 1970) [7].



Figure 11-22: Volodymyr Shcherbytsky [9].



Figure 11-23: Oleksiy Kosygin [10].



Figure 11-24: Anatoly Petrovich Alexandrov [12].



Figure 11-25: Yefim Slavsky [13].



Figure 11-26: Nikolai Dollezhal [14].

Location	Unit	First Power	MWe	Status
Lithuania				
Ignalina	1	1983	1185 (Originally 1300)	Closed 12/2004
Ignalina	2	1987	1185 (Originally 1300)	Closed 12/2009
Russia				
Kursk	1			

	2			
	3			
	4			
	5			
Leningrad	1			
	2			
	3			
	4			
Smolensk	1			
	2			
	3			
Ukraine				
Chernobyl	1			
	2			
	3			
	4			
	5			
	6			

Table 11-1: RBMK Reactors [15].

A Light Water Graphite-moderated Reactor (LWGR/RBMK)

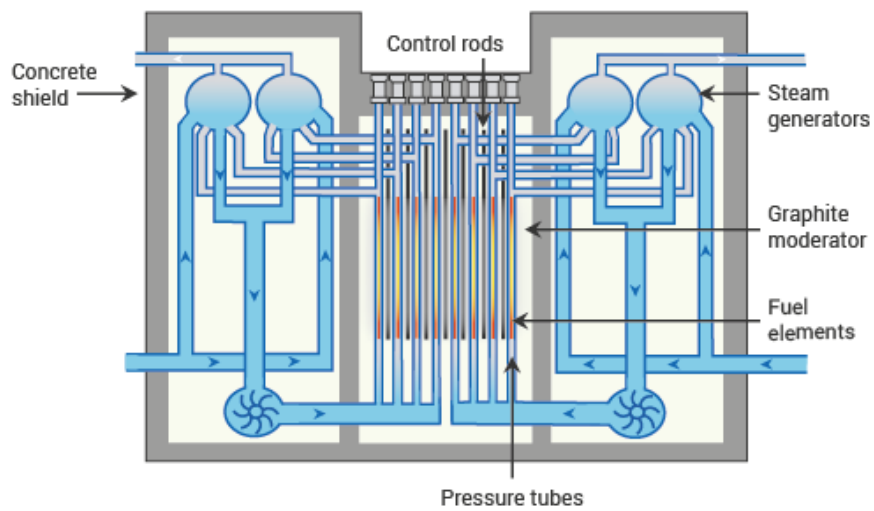


Figure 11-27: RBMK - LWGR [15].

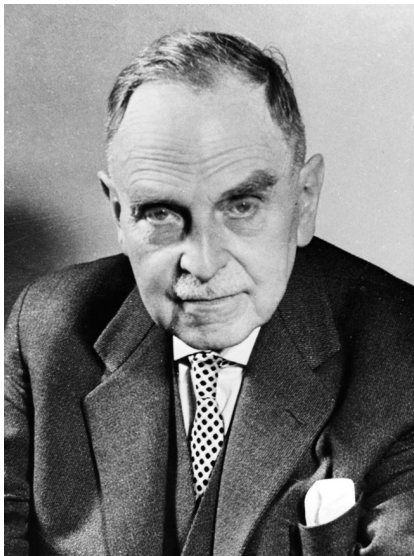


Figure 11-28: Otto Hann [16].



Figure 11-29: Fritz Strassmann [17].

PERIODIC TABLE OF ELEMENTS

Chemical Group Block

1100																
---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Figure 11-30: Periodic Table of Elements [18]

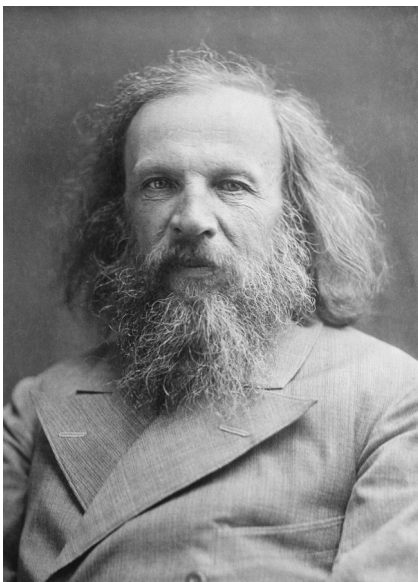


Figure 11-31: Dmitri Ivanovich Mendeleev [19]

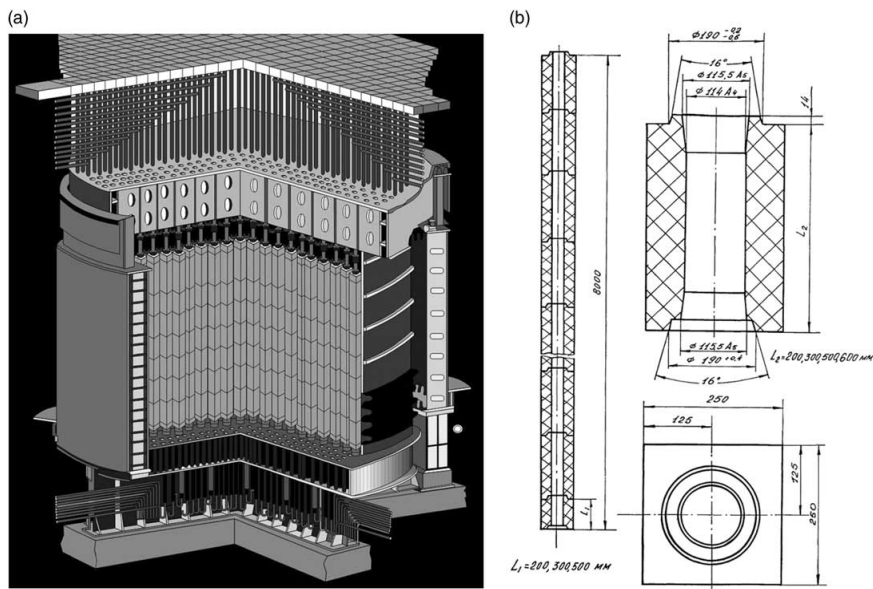


Figure 11-32: General view of the RBMK reactor [20].

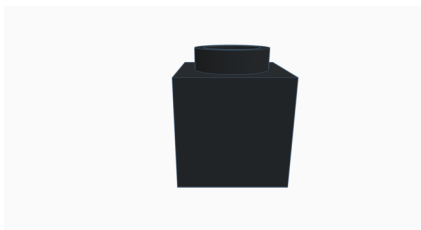


Figure 11-33: Graphite Block Depiction (Side View)

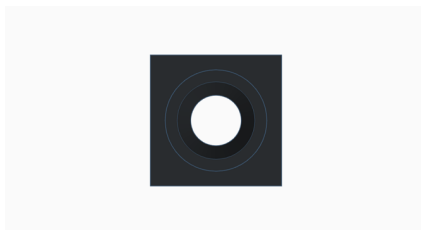


Figure 11-34: Graphite Block Depiction (Top Down)



Figure 11-35: Graphite Block Depiction (Bottom Angled)

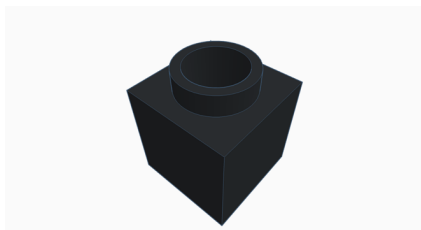


Figure 11-36: Graphite Block Depiction (Top Angled)

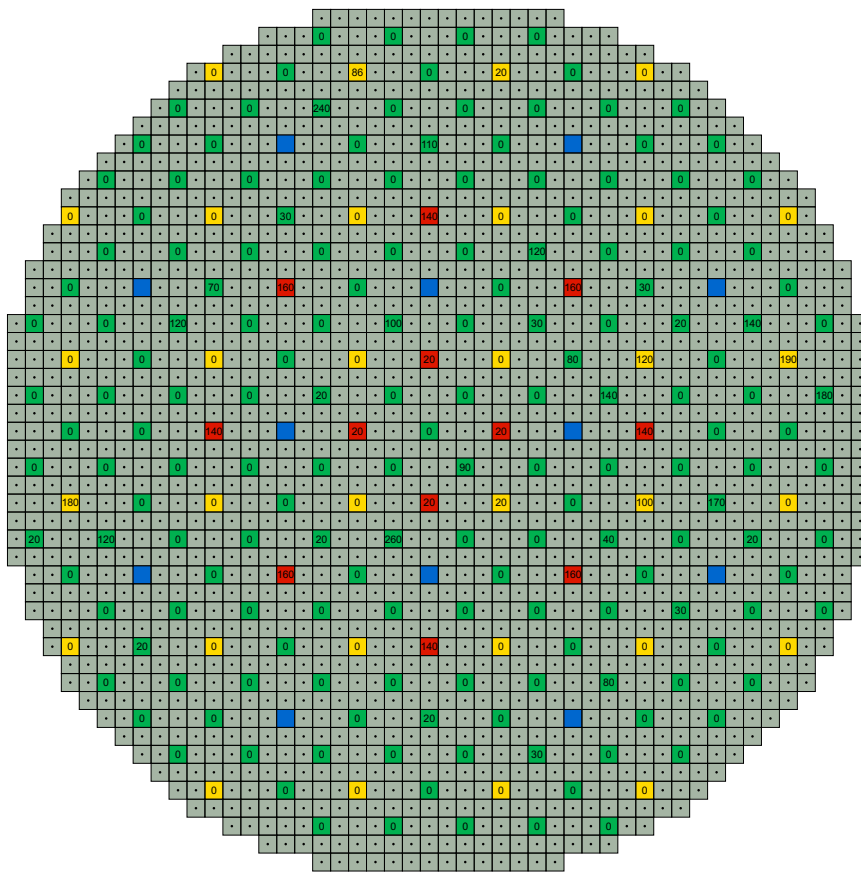


Figure 11-37: Schematic plan view of core layout, Chernobyl RBMK reactor No. 4. [21].



Figure 11-38: Families enjoying the Warm Sunny Day in Pripyat [22].



Figure 11-39: Leonid Fedorovich Toptunov [25].



Figure 11-40: Boris Stolyarchuk [26].



Figure 11-41: Aleksandr Fyodorovich Akimov [27]



Figure 11-42: Alexander (Sasha) Yuvchenko [28]

Chapter 4



Figure 11-43: Foss Therapies F812 Gamma Irradiator [2]



Figure 11-44: Irradiator Control Panel [3]



Figure 11-45: 3 Gamma Chamber Source Channels [4]



Figure 11-46: Inside the Gamma Chamber [5]

Figure 11-47: Arduino Rev 3 Front [6]

Figure 11-48: Arduino Rev 3 Back [6]

Chapter 5

Chapter 6

Chapter 7

Chapter 8